

Todd Takala

Lead Data Scientist, Machine Learning Engineer, AI/ML and Analytics

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Summary

Dynamic Data Scientist & Machine Learning Engineer with a proven track record in launching groundbreaking AI and ML data products, generating over \$1.3 billion in transformative business value. Expertise spans comprehensive end-to-end analytics, active stakeholder engagement, inspiring mentorship, and advanced MLOps.

Professional Experience

Lead Data Scientist - Caterpillar Inc. - Surprise, AZ

09/2022 to Present

- Architected and delivered end-to-end data products in Python, Snowflake, and R, including an AI platform that predicts equipment repair times. This initiative drove an \$80M reduction in warranty costs and earned a CEO Award
- Productionized models on Amazon SageMaker using deep learning (PyTorch, TensorFlow, Hugging Face), gradient boosting (XGBoost, LightGBM), and LLM/sentence-transformer embeddings; improved repair-time prediction accuracy by 40% and enhanced overall model performance
- Collaborated with senior leadership and stakeholders to facilitate an understanding of the project's broad impact and created precise user requirements. Ensured alignment with the business roadmap through effective written and verbal communication while championing the scaled agile framework

Reliability Analytics Manager - Amazon.com - Tempe, AZ

02/2022 to 09/2022

- Applied advanced analytics, mathematics, and statistics to develop a new KPI for autonomous robots utilized in warehouse automation through big data analysis. Authored a white paper on Weibull analysis for Design for Reliability, which led to its adoption by senior management
- Led a team of business intelligence engineers and data analysts to deploy a pipeline using Shiny R, improving real-time communication of robot status across all global fulfillment centers and enhancing operational efficiency and data analytics capabilities
- Utilized Amazon SageMaker for deep learning with PyTorch and TensorFlow. Employed XGBoost and LightGBM for gradient boosting to enhance model accuracy, significantly increasing predictive performance. Utilized AWS Redshift and PostgreSQL to enhance data management, accessibility, and processing speed
- Designed and implemented solutions utilizing large databases with Redshift and PostgreSQL. Supported data-driven decision-making and enhanced business intelligence capabilities

Data Engineering Manager - Empire-Cat - Mesa, AZ

01/2021 to 02/2022

- Develop a time-series AI model and MLOps pipeline using SQL, R, and Python for demand forecasting that identifies key supply chain risks
- Led a team to design, build, and maintain scalable data infrastructure, enhancing Empire's data-driven capabilities
- Developed a predictive ML model for engine reliability, durability, repair, preventive maintenance, and intervention, resulting in \$62 million in savings within 90 days

Reliability Engineering Manager - Empire-Cat - Mesa, AZ

01/2016 to 01/2021

- Supervised 8 engineers and technical communicators while serving as a customer-facing expert on mining equipment. Mentored up to 6 interns annually.
- Deployed reproducible research using R and SQL to track product issues and perform reliability analysis, establishing a template for Caterpillar's enterprise solution
- Led the root cause failure analysis program for Empire using a data-driven, fact-based, and evidence-oriented approach, supporting the FMEA and 6 Sigma processes at Caterpillar

- Applied the DMAIC Six Sigma methodology and computer science along with complex problem-solving skills to identify opportunities to improve operations, manufacturing, and maintenance of equipment fleets, yielding \$1.2 billion in savings over 6 years
- Led a team of 5 Technical Communicators in a client-facing role to effectively convey customer feedback to Caterpillar's engineering organizations, which drove improvements in product development and customer satisfaction
- Developed a condition monitoring application using R and SQL that enables fluid sample analysis on thousands of products, saving Empire \$100,000 in software licensing
- Collaborated with teams to design experiments that enhance reliability, safety, and durability
- Developed a Python and MS-SQL application to track performance and outcomes of prototype designs tested in the field

Certification & Technical Registration

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **Canonical:** Ubuntu Linux Professional Certificate
- **Data Camp:** AI Fundamentals Certificate • Certified SQL Associate, Data Scientist Professional Certificate
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IBM:** Advanced Deep Learning Specialist • RAG and Agentic AI Professional Certificate • Deep Learning with PyTorch, Keras and Tensorflow Professional Certificate
- **Komatsu North America:** Certified Technical Communicator
- **NCEES:** Passed FE Exam
- **US Department of Labor:** Operating Engineer Vocational Certificate

Education

Bachelor of Science in Mechanical Engineering - Arizona State University - Tempe, AZ 06/2003 to 05/2006
STEM, Multivariate Calculus, Differential Equations, Linear Algebra, Numerical Methods, Algorithms, Data Structures, Statistics, Mechanics, Material Science, Manufacturing, Finite Element Analysis, Computational Fluid Dynamics, Internal Combustion Engines, Mechanisms, Heat Transfer, Thermodynamics, Manufacturing, Economics, Cost, Finance, Control Systems, Design

Hard Skills

A/B Testing, Agentic AI, Agile Methodology, AI/ML, Anomaly Detection, Amazon Bedrock, Artificial Intelligence, AWS, BASH, CI/CD, Computer Science, RDBMS, Database Administrator, Data Science, DBT, Deep Learning, Design of Experiments, DevOps, Docker, Embedding, ETL, Fine Tuning, Advanced Excel, Financial Forecasting, Generative AI, Git, Linear Algebra, Large Language Models, LLM, Mathematics, Natural Language Processing, NLP, Pandas, Power BI, Public Cloud, Python, PyTorch, R, RAG, Research, Sagemaker, Scikit-learn, Scripting Language, SDLC, Simulation, SnapLogic, Snowflake, SQL Queries, Statistical Analysis, Streamlit, SQL, Tableau, TensorFlow, Transact SQL, Transformers, Version Control

Soft Skills

Adaptability, Attention to Detail, Business Requirements, Coaching, Communication Skills, Continuous Improvement, Creative Problem Solving, Data Driven, Decision making, Growth Mindset, Interviewing, Leadership Experience, Life Long Learning, Listening, Mentoring, Methodical, Multicultural Collaboration, Open minded, Problem Solving, Professional Services, Presentations, Public Speaking, Strategic Thinking, Team Building, Thought Leadership, Willing to Learn